An Exploration of STEM Attitudes of Female Students of Color Throughout Participation in Integrated STEM Curricula

University of Minnesota
Felicia Leammukda, Elizabeth Crotty, Jeanna Wieselmann, and Gillian Roehrig, Ph.D.

Purpose of the Study

- To investigate how select female students of color (FSOC) describe their success in STEM subjects
- To investigate the relationship between STEM attitudes and STEM proficiency among FSOC

Background and Context of Study

- Six urban middle school physical science classrooms in which engineering-based, integrated STEM units were implemented

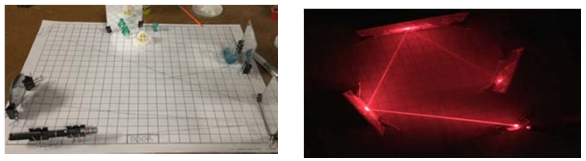


Figure 1. Example of prototype from the Laser Security System unit, one of the engineering-based integrated STEM units implemented in this study

Research Questions

1) *How do female students of color (FSOC) describe their success in STEM subjects?*
2) *What is the relationship between STEM attitudes and STEM proficiency among female students of color?*

Data & Participants

Quantitative data:
87 FSOC took a 28-item Likert-scale STEM Attitude Survey and a 30-item multiple choice assessment of math, science, and engineering after implementation of engineering-based integrated STEM unit

Qualitative data:
Semi-structured interviews with twelve target students, all FSOC, from four classrooms were conducted after implementation of integrated STEM unit

Methodology

Convergent parallel mixed methods study in which qualitative and quantitative data were collected in parallel after implementation of STEM unit

Analysis

- Based on results of STEM attitude survey and STEM proficiency assessment, students were given one of four Johari Window Assignments -- High Aligned, Low Aligned, Overconfident, or Under confident
- Semi-structured interviews with target students further contribute to the understanding of how FSOC view their STEM success

This research was made possible by the National Science Foundation grant #DRL-1238140. The findings, conclusions, and opinions herein represent the views of the author and do not necessarily represent the view of personnel affiliated with the National Science Foundation.

NSF # 1238140

Findings

Johari Window Assignment (JA)

	High Self-Perception (POSITIVE STEM ATTITUDE)	Low Self-Perception (NEGATIVE STEM ATTITUDE)
High Performance (STEM PROFICIENT)	ALIGNED (high) 5 (5.8%) *Amaya, Dawn*	MISALIGNED (under confident) 6 (6.9%) *(None)*
Low Performance (NOT STEM PROFICIENT)	MISALIGNED (overconfident) 41 (47.1%) *Gladys, Allie, Bea, Melissa, Clarissa*	ALIGNED (low) 35 (40.2%) *Karen, Erin, Maria, Ava*

Figure 2. Modified Johari Window*. The number and percentage of FSOC within each Johari Window assignment (JA) taken from the total 87 participants is noted. Pseudonyms of target students shown in italics.

*Adapted from Luft, J. (1961). The Johari Window: A Graphic Model of Awareness in Interpersonal Relations. *Human Relations Training News, 5* (1) 6-7.

How Target Students Describe Their Success in STEM Subjects: Semi-Structured Interviews

# Target Students (out of 12)	Metrics of Success
11	Grades/scores
6	Completion of work
6	Learning new things
4	Peer comparison
3	Teacher evaluation
3	Collaboration
3	Speed
2	Ease of understanding
2	Leadership/helping peers
1	Mistakes as an opportunity to learn
1	Effort-based/hard work

There is a high percentage FSOC who are overconfident in STEM (positive STEM attitude, but not STEM proficient).

- 12.7% of FSOC in this study are proficient in STEM
- 52.9% have a positive STEM attitude
- 47.1% are overconfident (positive STEM attitude and not STEM proficient)

Target Students' quotes from Semi-structured Interviews:

Gladys' metrics of success include teacher evaluation --
"I felt accomplished...there was this paper that we had to do… at the end [my teacher] told me that I did really good because I paid attention and I wasn't… playing around… Yeah, I felt accomplished."
-Gladys (Overconfident)

Clarissa believes that success comes from hard work. Her family came to the U.S. so that she could get a good education --
"[My] goal is to…work really hard and get [a] scholarship. I…had to study really hard to pass it. So I always study."
-Clarissa (Overconfident)

Melissa's metrics of success includes completion of work --
"The time I felt successful is when I got all my math work done."
-Melissa (Overconfident)

Conclusions

Female students of color with positive STEM attitudes are not necessarily STEM proficient. Our FSOC target students in this study do not view their STEM success by proficiency (grades/scores) alone. Interviews with target students reveal that completion of work, learning new things, peer comparison, teacher evaluation, and hard work are also metrics of success when it comes to STEM.

Implications

- We may be creating a culture of overconfidence among our students
- Students need to know their current limitations in order to truly overcome barriers to STEM success
- In sixth grade, students may be interested in STEM, but if they are overconfident they may have a more difficult time overcoming STEM barriers
- Challenge for educators: How to appropriately support female students of color in their success in STEM and foster self-efficacy

Engineering to Transform the Education of Analysis, Measurement, and Science